TOPIC:-

Obtain information about commonly used gadgets or devices which are based on the principles of Newton:-

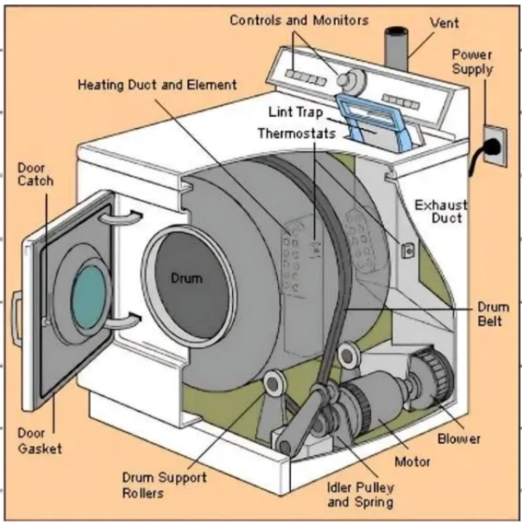
Introduction:-

Well, we all know that Newton has given some laws related to motion. In present time we use various gadgets, but some of them follow the laws of Newton.With the help of this presentation, we will try to see which devices are and how they follow the laws of Newton.

Sir Isaac Newton discovered that Newton's laws of motion describe how an object's motion changes when it is acted upon by a force.Newton's second law of motion defines that the force on an object is equal to its mass times its acceleration. Newton's third law of motion is defined as "for every action, there is an equal and opposite reaction." Sometimes we even recognize third law of motion as law of action-reaction.

• Let's see the examples of devices working on the principle of Newton's law.

1) Washing Machine Dryer:-



A washing machine dryer entirely works upon the principle of inertia. To dry the clothes, the drum of the washing machine dryer is subjected to motion, which further causes the clothes to move. However, the water molecules contained in the cloth do not follow the motion and stay at their position of rest. Due to the gravitational pull of the earth, the water gets collected at the base of the drum. The holes of the drum let the water out, leaving the clothes dry.

2) Brakes applied by a Bus Driver abruptly:-

While travelling on a bus, when the bus driver abruptly applies the brakes, we tend to fall momentarily in the forward direction. The reason behind this jerk felt by the passengers sitting inside the bus is due to inertia. Due to the inertia of motion, our body continues to maintain a state of motion even after the bus stops, thereby pushing us in the forward direction.

Let's know more with separate examples with example with respect to the laws

Newton's First Law of Motion based devices

1) Fidget Spinner:-

The object will most likely not start spinning on itself before turning, but once turned, if no external force is provided, it will spin indefinitely.The electric fan continues to move for a period after the electricity is turned off.

** Based on / follows the Newton's second law of motion**

1) Airbags:-

In a forward motion, airbags mitigate the effect of force

2)Seat belts:-

A seat belt restricts the motion of our upper body so that we do not knock our heads when the automobile stops.

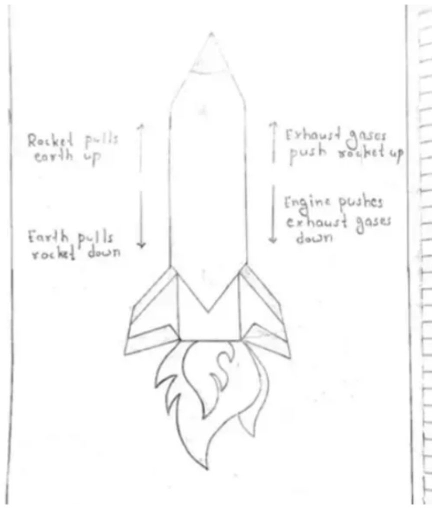
Newton's Third Law of Motion

Objects follow the principle of Newton's Third Law of Motion

1) firing of bullet:-

When a gun is fired, the gun puts a force on the bullet that propels it forward.

2) Principle of Rocket Propulsion:-



The propulsion system purpose is to generate thrust. Thrust is the force that moves a rocket through the air and through space.The rocket engine pushes the rocket upward by releasing hot burning fuel in a downward direction, while the gases exert an equal and opposite force.

Conclusion:-

In this way, the working of a rocket is based on the principles of Newton's Third Law of Motion